

Prevalence of potentially neuropathic Campylobacter jejuni strains on commercial broiler chicken products.

Campylobacteriosis is the most common antecedent infection leading to the development of inflammatory neuropathies including Guillain Barré syndrome (GBS) and Miller Fisher syndrome (MFS), with alterations in surface proteins and genetic polymorphisms conferring increased risk. Poultry is the most common source of *C. jejuni* infection in industrialized countries, including the US. There are no data on the prevalence on consumer poultry products of various strains of *C. jejuni*, including those hypothesized to be associated with neuropathy. To study this, *C. jejuni* was isolated from fresh broiler chicken products purchased from grocery stores in the Baltimore area. LOS subtypes and specific genetic polymorphisms were determined by PCR and DNA sequencing. The observed relative proportions of LOS subtypes and genetic polymorphisms in the *cstII* gene (encoding bacterial sialyltransferases involved in LOS synthesis in *C. jejuni*) were characterized and compared to those reported in published studies of patients with GBS, MFS and uncomplicated enteritis. Commercial poultry products carry a relatively high prevalence of *C. jejuni* strains that have been associated with neuropathic sequelae. The relative proportions of LOS classes in poultry isolates were similar to those reported in isolates from human enteritis cases, and in some instances also similar to isolates from patients diagnosed with neuropathic disease. In terms of *cstII* polymorphisms, there were also similarities between isolates from poultry and those from patients with GBS and MFS. Copyright © 2010 Elsevier B.V. All rights reserved.